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Bihar Engineering University, Patna

B.Tech 3rd Semester Special Exam., 2020

Time: 03 Hours

Code: ESC-202 (101302) *Subject: Basic Electronics* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer/Fill in the blanks/Write true or false (any seven): **[2x7=14]**

(a) Why is filter required to be added after rectifier circuit?

(b) What is the application of voltage regulator?

(c) DC value of pure AC waveform is _____.

(d) Which of the transistor currents is always the largest?

(e) What is the source of leakage current in a transistor?

(f) What is the major difference between a bipolar and a unipolar device?

(g) Give the advantage of negative feedback in amplifier.

(h) BUT amplifier changes the frequency and shape of the signal.

(i) CMMR for ideal operational amplifier is _____.

(j) Slew rate for ideal operational amplifier is _____.

Q.2 Solve both questions : **[6+8=14]**

(a) Draw volt-ampere characteristics of a Zener diode. What is meant by the knee voltage in the curve?

(b) Describe in detail the avalanche breakdown mechanism in a diode.

Q.3 Solve both questions : **[6+8=14]**

(a) Explain the principle of operation of LED with the help of diagram.

(b) Derive the expression for I_C in terms of I_E and α_{dc} for a common-base BUT configuration. For a given α_{dc} of 0.998, determine I_C, if I_E= 4mA.

Q.4 Solve both questions : **[5+9=14]**

(a) Define the stability factor. What does smaller stability factor indicate?

(b) Derive the expression for stability factor for voltage divider bias (self-bias) circuit.

Q.5 Solve both questions : **[7+7=14]**

(a) Explain the construction and characteristics of FET.

(b) Draw and explain h-parameter small signal model for BJT.

Q.6 Solve both questions : **[8+6=14]**

(a) Explain the working of common emitter amplifier with the help of circuit diagram.

(b) What are the roles of coupling and bypass capacitors? Explain in detail.

Q.7 Solve both questions : **[6+8=14]**

(a) Describe the RC phase-shift oscillator with circuit diagram.

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(b) Explain the working of high-frequency LC oscillator with the help of circuit diagram.

Q.8 Solve both questions : **[7+7=14]**

(a) Explain the working of voltage series feedback amplifier with the help of block diagram.

(b) Design an integrator using op-amp and explain its working.

Q.9 Solve both questions : **[7+7=14]**

(a) Write the circuit diagram of differential amplifier configurations and explain its working.

(b) Explain about CMRR, slew rate and concept of virtual ground.

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